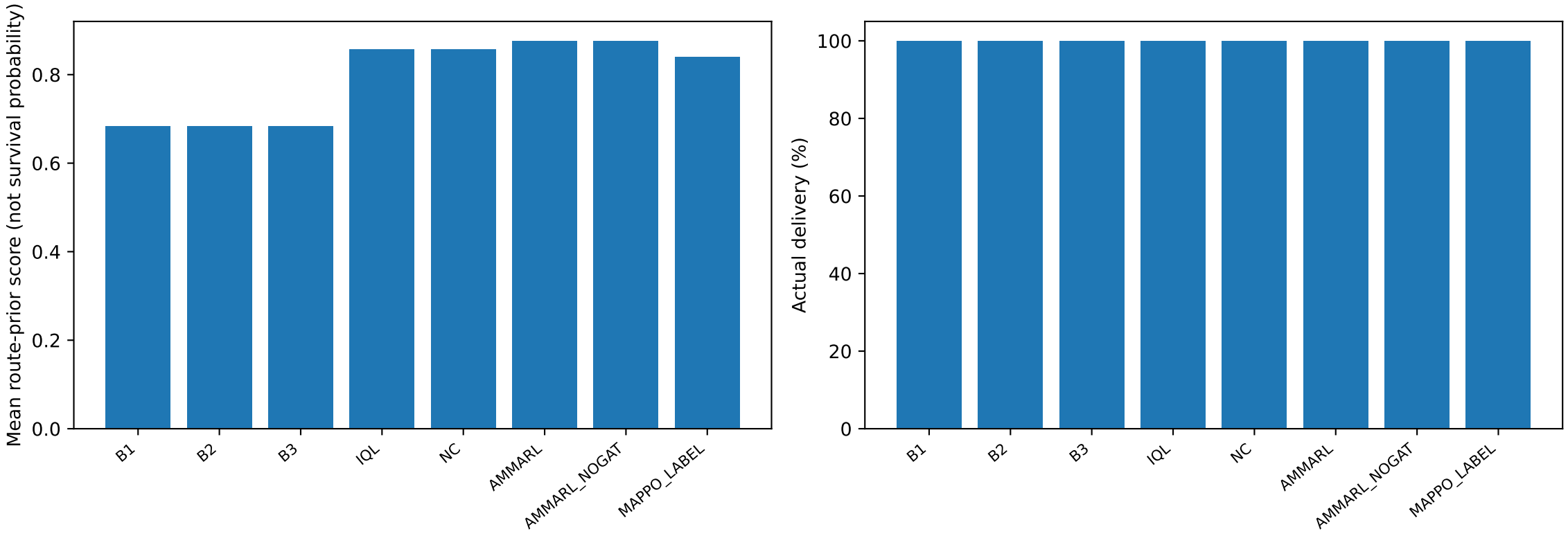
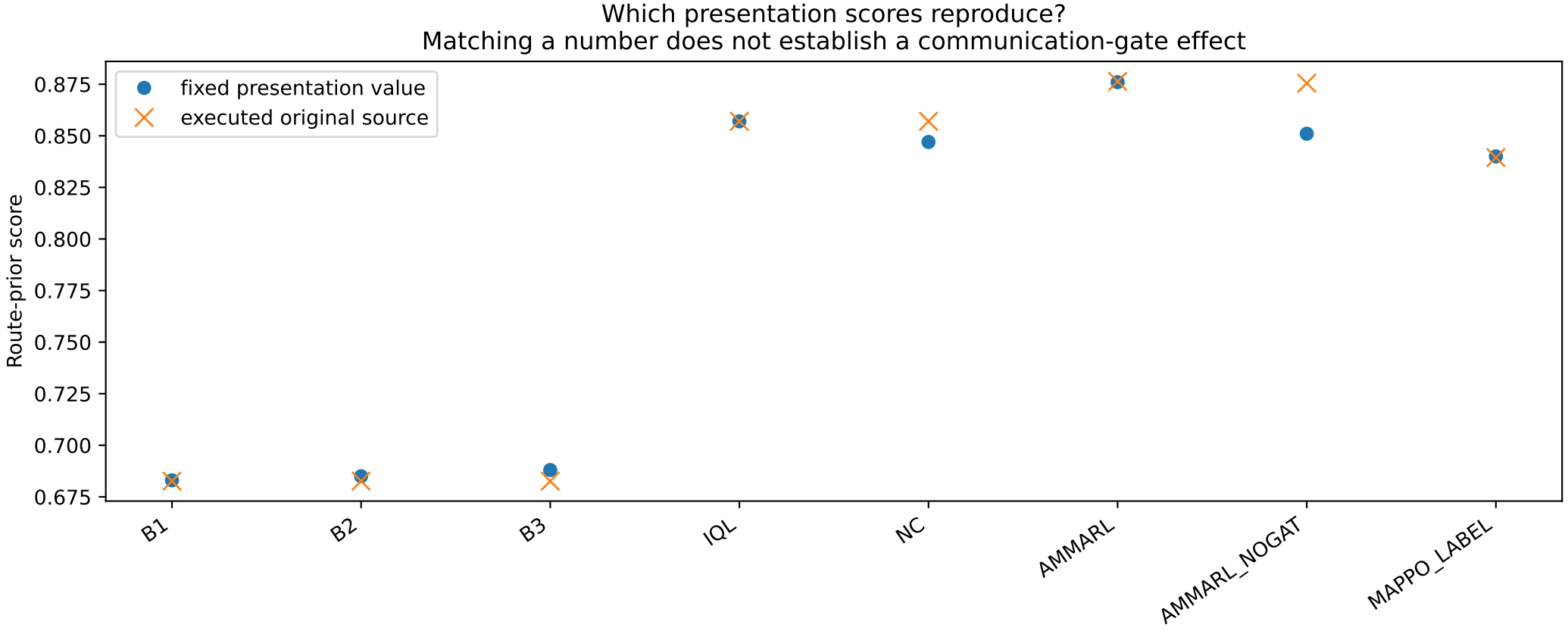
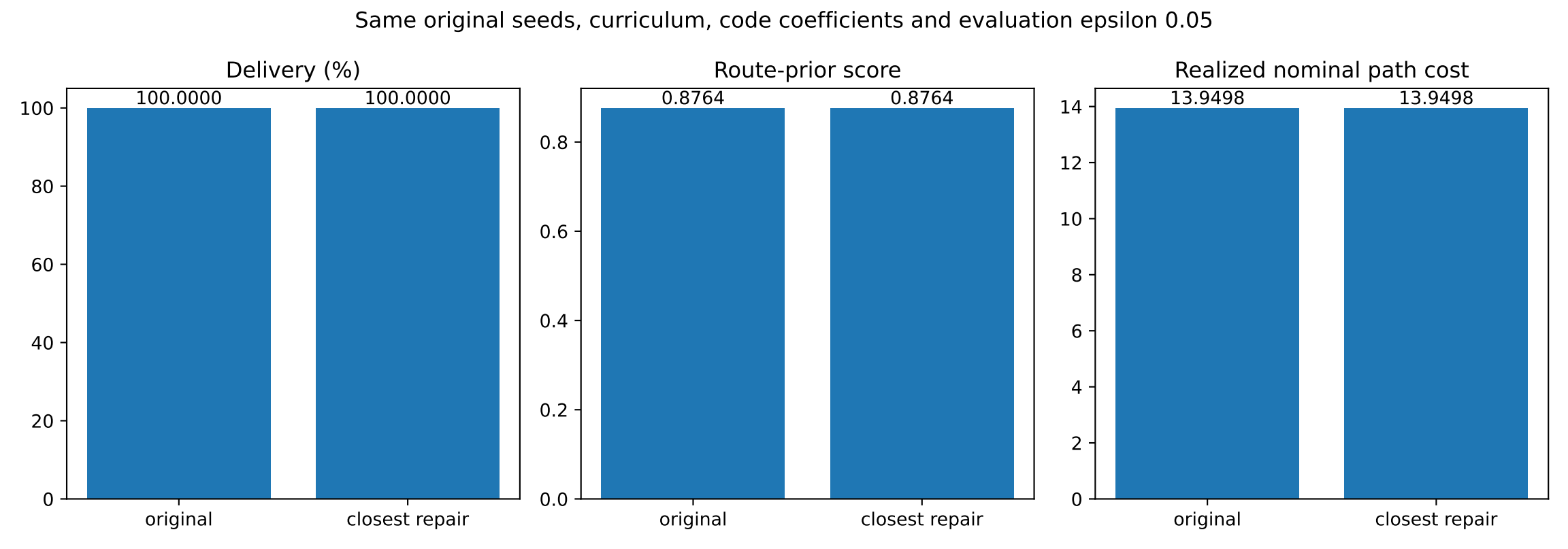
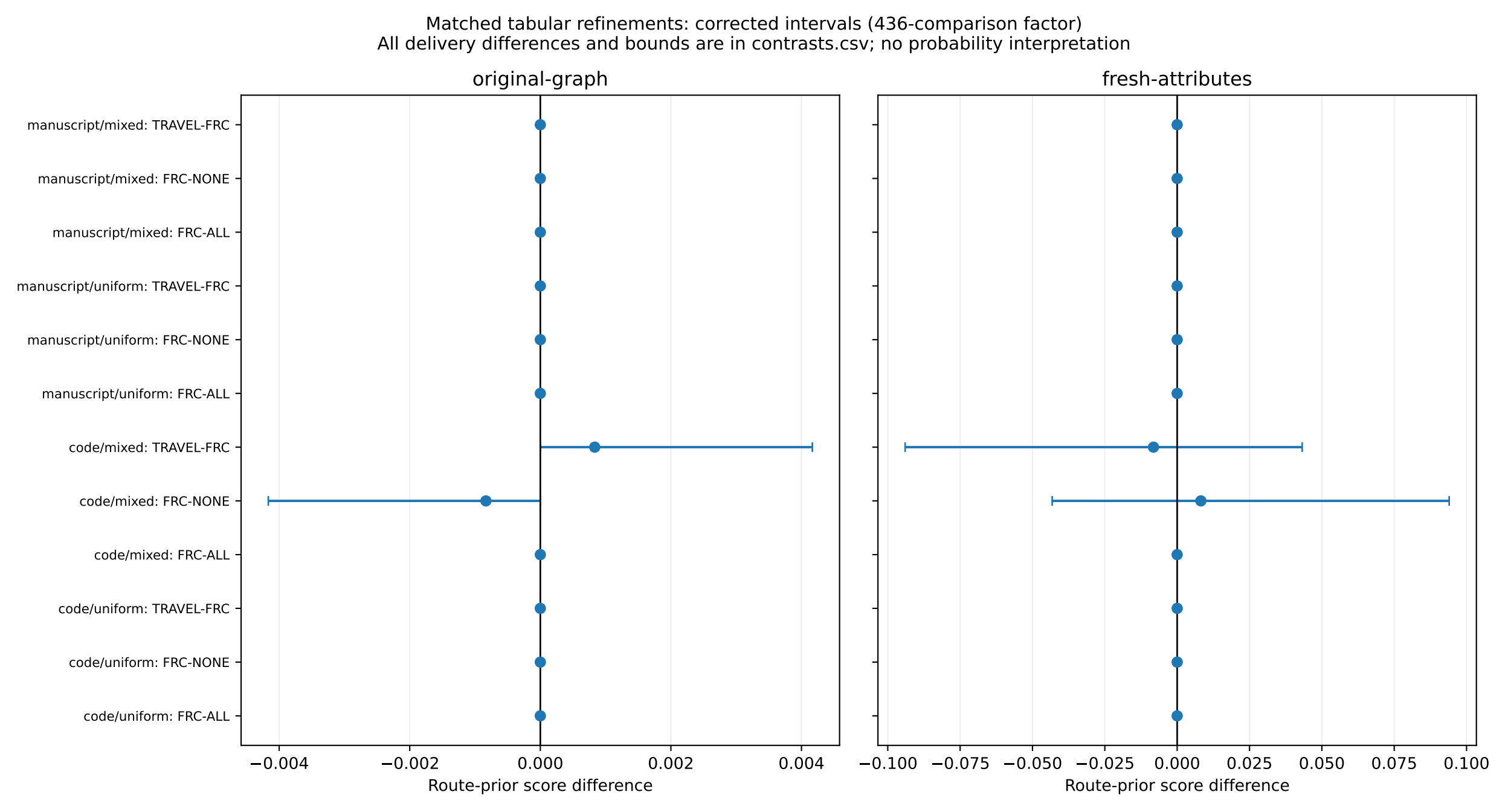
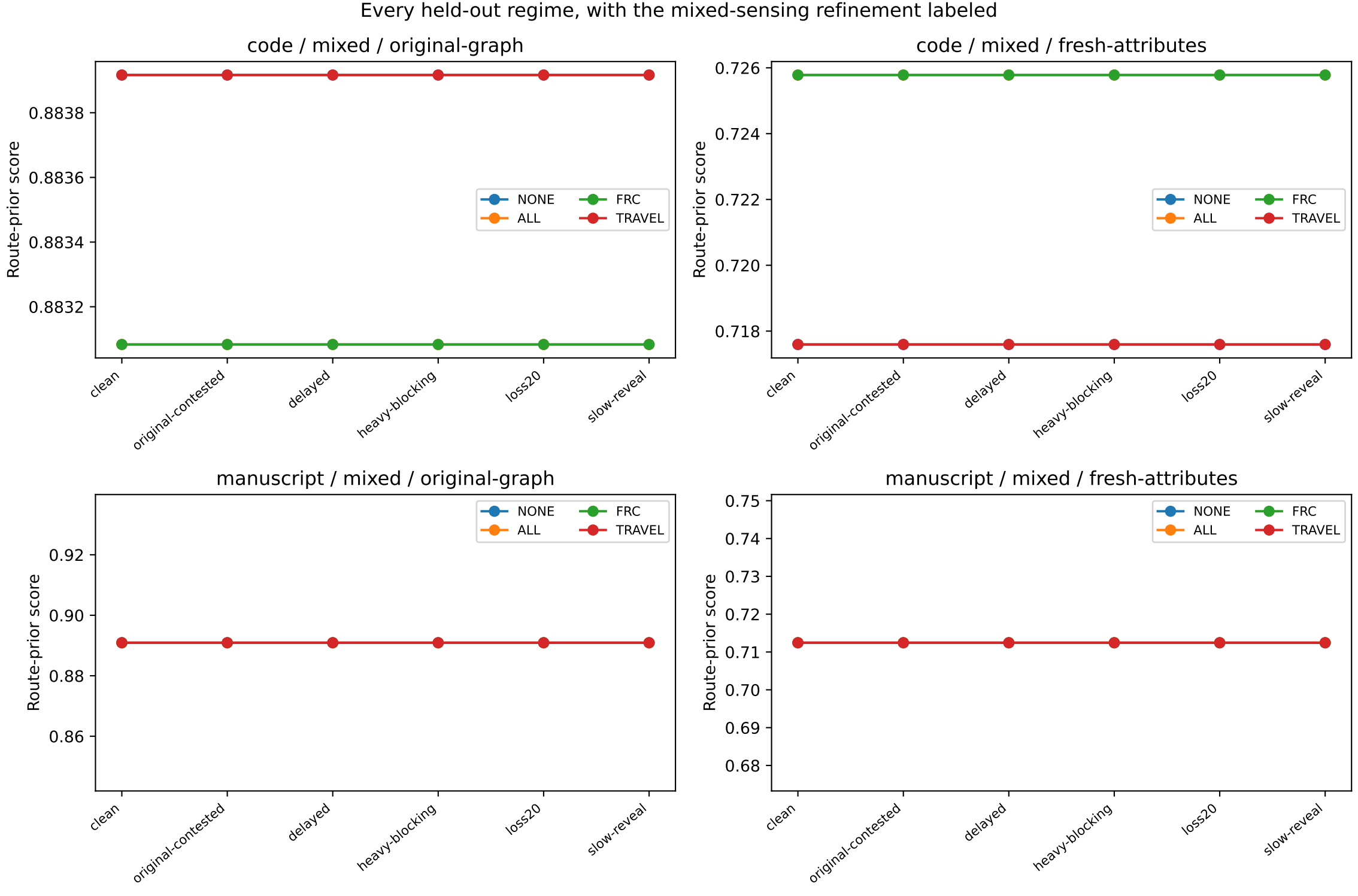
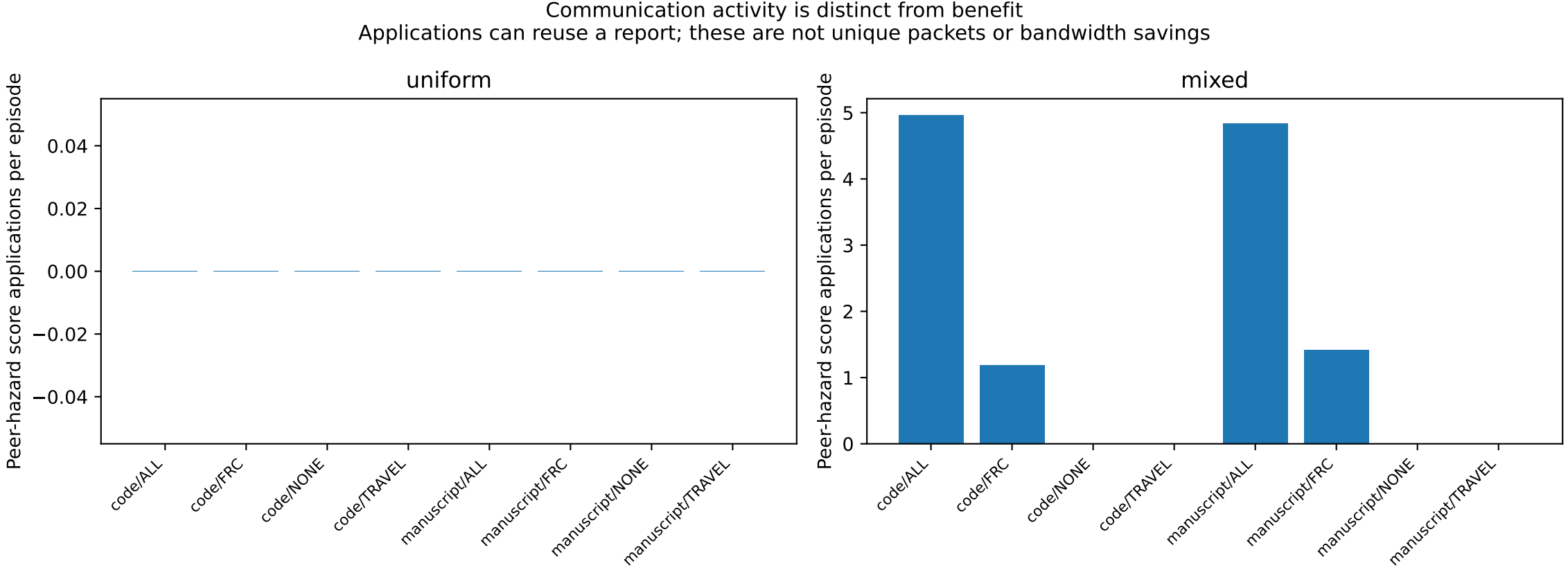

Unmodified submitted source: original M4 protocol
AMMARL hazard-buffer entries and hazard uses =0
Mean route-prior score (not survival probability)
0.0
0.2
0.4
0.6
0.8
B1
B2
B3
IQL
NC
AMMARL
AMMARL_NOGAT
MAPPO_LABEL
Actual delivery (%)
0
20
40
60
80
100
B1
B2
B3
IQL
NC
AMMARL
AMMARL_NOGAT
MAPPO_LABEL

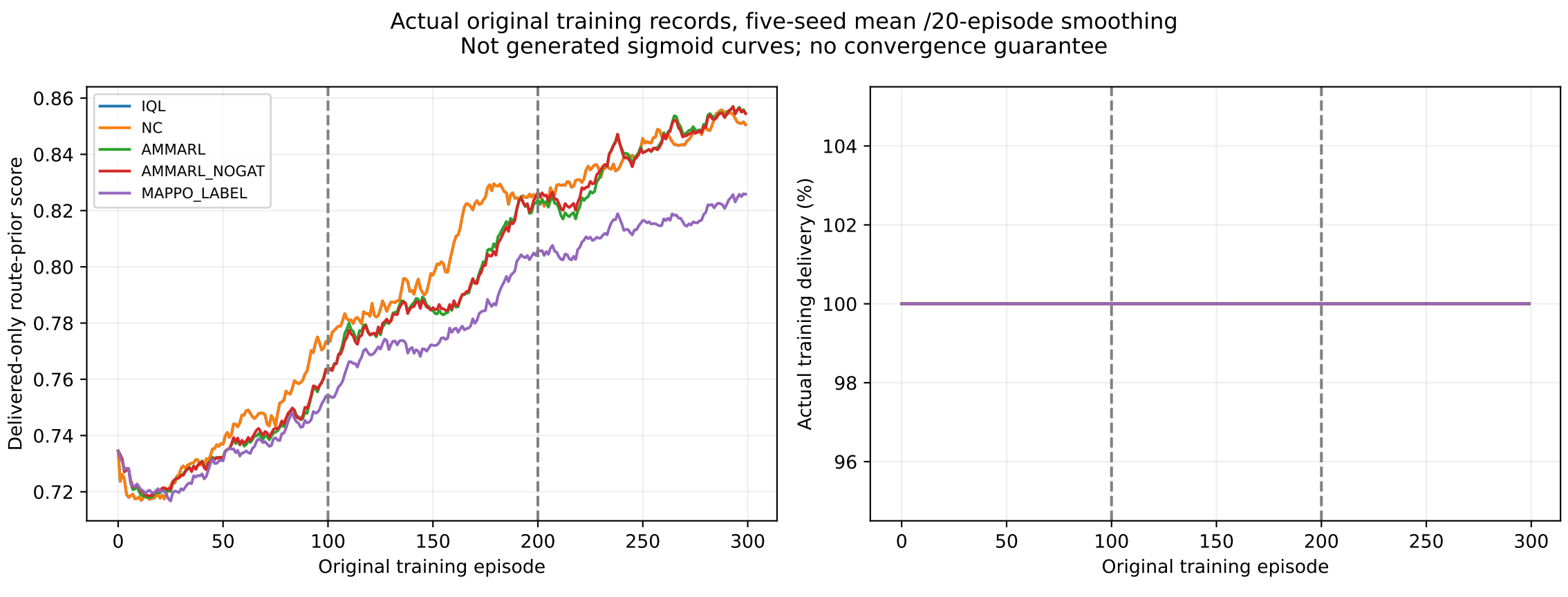

Which presentation scores reproduce?
Matching a number does not establish a communication-gate effect
fixed presentation value
executed original source
Route-prior score
0.875
0.850
0.825
0.800
0.775
0.750
0.725
0.700
0.675
B1
B2
B3
IQL
NC
AMMARL
AMMARL_NOGAT
MAPPO_LABEL

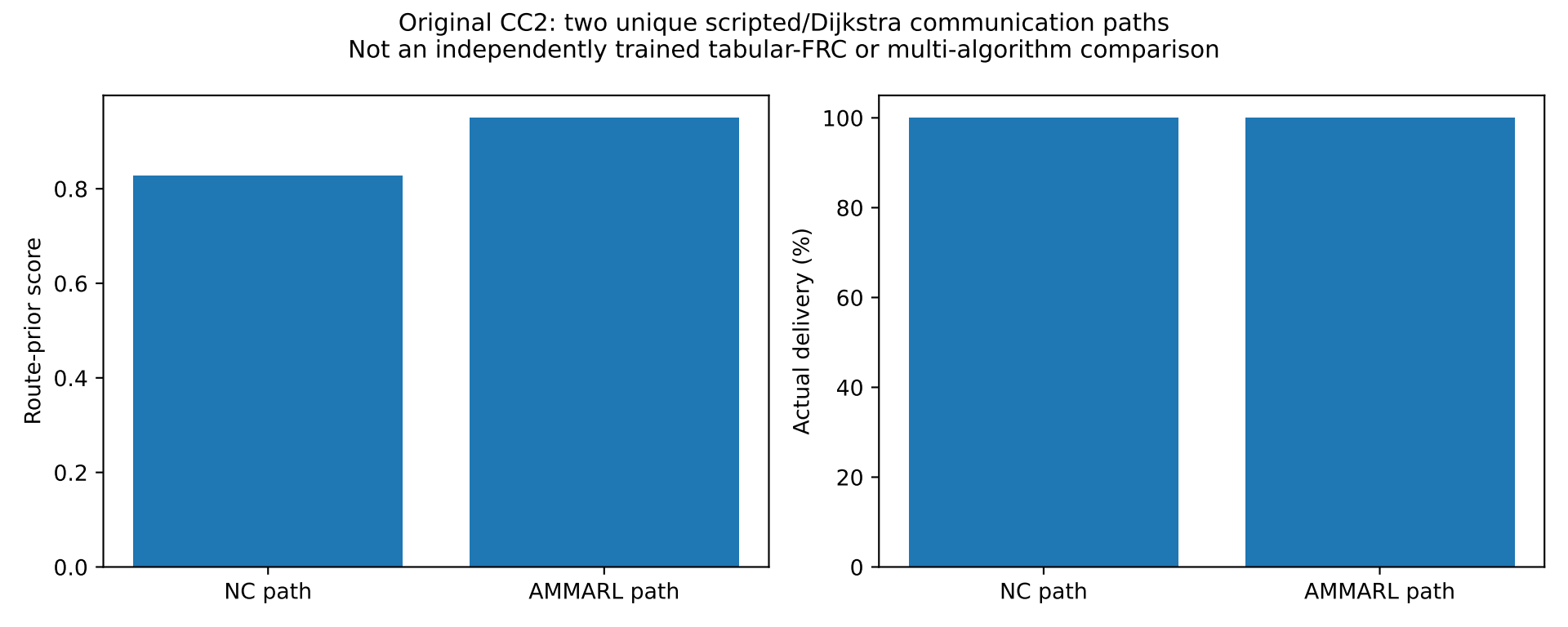

Same original seeds, curriculum, code coefficients and evaluation epsilon 0.05
Delivery (%)
100.0000
100.0000
100
80
60
40
20
0
original
closest repair
Route-prior score
0.8764
0.8764
0.8
0.6
0.4
0.2
0.0
original
closest repair
Realized nominal path cost
13.9498
13.9498
14
12
10
8
6
4
2
0
original
closest repair


Matched tabular refinements: corrected intervals (436-comparison factor)
All delivery differences and bounds are in contrasts.csv; no probability interpretation
original-graph
fresh-attributes
manuscript/mixed: TRAVEL-FRC
manuscript/mixed: FRC-NONE
manuscript/mixed: FRC-ALL
manuscript/uniform: TRAVEL-FRC
manuscript/uniform: FRC-NONE
manuscript/uniform: FRC-ALL
code/mixed: TRAVEL-FRC
code/mixed: FRC-NONE
code/mixed: FRC-ALL
code/uniform: TRAVEL-FRC
code/uniform: FRC-NONE
code/uniform: FRC-ALL
−0.004
−0.002
0.000
0.002
0.004
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
0.100
Route-prior score difference
Route-prior score difference


Every held-out regime, with the mixed-sensing refinement labeled
code / mixed / original-graph
code / mixed / fresh-attributes
manuscript / mixed / original-graph
manuscript / mixed / fresh-attributes
Route-prior score
NONE
ALL
FRC
TRAVEL
clean
original-contested
delayed
heavy-blocking
loss20
slow-reveal
0.8838
0.8836
0.8834
0.8832
0.726
0.724
0.722
0.720
0.718
0.92
0.90
0.88
0.86
0.75
0.74
0.73
0.72
0.71
0.70
0.69
0.68


Communication activity is distinct from benefit
Applications can reuse a report; these are not unique packets or bandwidth savings
uniform
mixed
Peer-hazard score applications per episode
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
code/ALL
code/FRC
code/NONE
code/TRAVEL
manuscript/ALL
manuscript/FRC
manuscript/NONE
manuscript/TRAVEL


Actual original training records, five-seed mean /20-episode smoothing
Not generated sigmoid curves; no convergence guarantee
IQL
NC
AMMARL
AMMARL_NOGAT
MAPPO_LABEL
Delivered-only route-prior score
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0
50
100
150
200
250
300
Original training episode
Actual training delivery (%)
104
102
100
98
96
0
50
100
150
200
250
300
Original training episode


Original CC2: two unique scripted/Dijkstra communication paths
Not an independently trained tabular-FRC or multi-algorithm comparison
Route-prior score
0.0
0.2
0.4
0.6
0.8
NC path
AMMARL path
Actual delivery (%)
0
20
40
60
80
100
NC path
AMMARL path